

Yuewen Gao

Nanjing University, B.S. in Intelligent Science and Technology

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Education

B.S. in Intelligent Science and Technology

Nanjing University, Nanjing, China

Sep 2023 - Jun 2027

(Expected)

- **GPA:** 4.44/5.00

Research Interest

- **Human-Centric AI:** Investigating adaptive agents that dynamically align with human cognitive architectures and value systems, fostering mutual understanding while safeguarding human agency in open-ended tasks.
- **Visualization and Intelligent Exploration:** Creating scalable and AI-augmented visualization systems to facilitate progressive knowledge discovery from multimodal data.
- **Cognitive Foundations of Interaction:** Grounding human-system interaction in empirical cognitive theories to derive foundational design principles for next-generation interfaces.

Research Experience

Research Assistant, Northeastern Human-Centered AI Lab

Jul 2026 - Present

Advised by Prof. Dakuo Wang and Bingsheng (Arthur) Yao

- **Human-Agent Intent Co-construction (Ongoing):** Lead a theory-informed project investigating how AI agents establish, externalize, and repair their interpretations of evolving human intent. Develop a coding framework grounded in communication theory to analyze collaboration episodes and translate observed breakdowns into agent interaction mechanisms.
- **Collaborative Human-AI Systems (Under Review):** Contributed to interviews and qualitative coding for two studies examining how shareable and continuable human-LLM conversation trajectories support team deliberation [4], and how trajectory-centered interfaces enable users to coordinate multiple AI agents in knowledge work [5].

Research Assistant, iMate Lab, Nanjing University

Feb 2025 - Present

Advised by Prof. Ke Xu

- **Non-intrusive Assistance for Sketch-Based Ideation (Under Review):** Led the project from problem formulation and system design through full-stack implementation and mixed-method evaluation. Developed a sketch-based proactive agent, informed by flow theory, that coordinates whether, when, and what support to provide, and examined users' perceptions of intervention timing and relevance [3].
- **Multimodal Visualization for News-Video Sensemaking (CCF CAD/CG 2026):** Led the design and implementation of an event-centered visual analytics system for serial news videos. Developed the multimodal processing and event-aggregation pipeline, implemented interactive exploration views, and conducted user research on event-centered video sensemaking [2].
- **Adaptive Recommendations for Network Exploration (ACM CHI 2026):** Designed and implemented an exploration-provenance module that records users' analysis paths and connects them with adaptive recommendations for progressive network exploration. Contributed to the literature review, core system development, and manuscript writing [1].

Publications & Manuscripts

- [1] Wenchao Li, **Yuewen Gao**, Yu He, Cong Zhu, and Ke Xu. “NetworkCanvas: Supporting Progressive Network Visualization Exploration via Adaptive Recommendations.” *ACM CHI 2026*.
- [2] **Yuewen Gao**, Yu He, Xianglei Lv, Yingying Dong, and Ke Xu. “Multimodal Analysis of News Videos: Event-Based Visual Summarization over Serial Broadcasts.” *CCF CAD/CG 2026*.
- [3] **Yuewen Gao**, Wenchao Li, Yiqing Yang, Lejia Li, Yutong Wu, and Ke Xu. “ProactiveFlow: Coordinating Proactive AI Support for Sketch-Based Ideation.” *Under Review, ACM CHI 2027*.
- [4] Jiaju Chen, Jingfei Huang, Chaoran Chen, **Yuewen Gao**, et al. “Carrying Thoughts Forward: Exploring How Shareable and Continuable Human–LLM Conversation Trajectories Reshape Team Deliberation.” *Under Review, ACM CHI 2027*.
- [5] Menglin Zhao, Fangrui Sima, Yuxuan Lu, **Yuewen Gao**, et al. “Trajectory-Centered Interaction: User-Directed Coordination of Multiple AI Agents in Knowledge Work.” *Under Review, ACM CHI 2027*.

Awards & Honors

People’s Scholarship (University-level Merit Scholarship)	Nov 2024, 2025
Honorable Mention, Mathematical Contest in Modeling (MCM)	Jan 2025

Technical Skills

Programming Languages: Python, TypeScript, JavaScript, C/C++, LaTeX

Libraries/Platforms: React, Node.js, PyTorch, D3.js, G6.js, ECharts, Git, Docker

Languages: Chinese (Native); English (TOEFL iBT 105, CET-4 658, CET-6 640)